Online Resource 1

Article: Exercise versus or in addition to pembrolizumab with neoadjuvant chemotherapy for early-stage triple-negative breast cancer: a model-based cost-effectiveness analysis in Japan

Journal: Breast Cancer Research and Treatment

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Reference numbers refer to the reference list of the main article.

Supplementary Methods S1. Literature searches, survival model, validation and probabilistic sampling

Literature searches

The statement in the Introduction that no economic evaluation of exercise during neoadjuvant therapy for breast cancer was identified rests on three searches, run on 26 October 2026 in PubMed (title and abstract words and Medical Subject Headings) and Europe PMC (title and abstract words), each limited to publication years 2000 to 2026. The first search combined exercise, neoadjuvant or preoperative treatment, breast cancer and economic evaluation. The second searched cancer of any type with neoadjuvant treatment, in any spelling, or with the phrases during treatment and during oncological treatment. The third searched breast cancer with chemotherapy in any setting. The title and abstract of every record were read. The search strings were as follows.

First search, PubMed: (exercise[tiab] OR "physical activity"[tiab] OR "Exercise"[mh] OR "Exercise Therapy"[mh] OR prehabilitation[tiab] OR "aerobic training"[tiab] OR "resistance training"[tiab]) AND (neoadjuvant[tiab] OR preoperative[tiab] OR "Neoadjuvant Therapy"[mh]) AND ("Breast Neoplasms"[mh] OR "breast cancer"[tiab] OR "breast carcinoma"[tiab]) AND ("cost-effectiveness"[tiab] OR "cost effectiveness"[tiab] OR "cost-utility"[tiab] OR "cost utility"[tiab] OR "economic evaluation"[tiab] OR QALY*[tiab] OR "Cost-Benefit Analysis"[mh]) AND ("2000/01/01"[dp] : "2026/12/31"[dp])

First search, Europe PMC: ((TITLE:exercise OR ABSTRACT:exercise) OR (TITLE:"physical activity" OR ABSTRACT:"physical activity") OR (TITLE:prehabilitation OR ABSTRACT:prehabilitation) OR (TITLE:"aerobic training" OR ABSTRACT:"aerobic training") OR (TITLE:"resistance training" OR ABSTRACT:"resistance training")) AND ((TITLE:neoadjuvant OR ABSTRACT:neoadjuvant) OR (TITLE:preoperative OR ABSTRACT:preoperative)) AND ((TITLE:"breast cancer" OR ABSTRACT:"breast cancer") OR (TITLE:"breast neoplasms" OR ABSTRACT:"breast neoplasms") OR (TITLE:"breast carcinoma" OR ABSTRACT:"breast carcinoma")) AND ((TITLE:"cost-effectiveness" OR ABSTRACT:"cost-effectiveness") OR (TITLE:"cost effectiveness" OR ABSTRACT:"cost effectiveness") OR (TITLE:"cost-utility" OR ABSTRACT:"cost-utility") OR (TITLE:"cost utility" OR ABSTRACT:"cost utility") OR (TITLE:"economic evaluation" OR ABSTRACT:"economic evaluation") OR (TITLE:QALY OR ABSTRACT:QALY) OR (TITLE:"quality-adjusted life year" OR ABSTRACT:"quality-adjusted life year")) AND PUB_YEAR:[2000 TO 2026]

Second search, PubMed: (exercise[tiab] OR "physical activity"[tiab] OR "Exercise"[mh] OR "Exercise Therapy"[mh] OR prehabilitation[tiab] OR "aerobic training"[tiab] OR "resistance training"[tiab]) AND (cancer[tiab] OR "Neoplasms"[mh]) AND ("neo adjuvant"[tiab] OR neo-adjuvant[tiab] OR "(neo)adjuvant"[tiab] OR neoadjuvant[tiab] OR "during treatment"[tiab] OR "during oncological treatment"[tiab]) AND ("cost-effectiveness"[tiab] OR "cost effectiveness"[tiab] OR "cost-utility"[tiab] OR "cost utility"[tiab] OR "economic evaluation"[tiab] OR QALY*[tiab] OR "Cost-Benefit Analysis"[mh]) AND ("2000/01/01"[dp] : "2026/12/31"[dp])

Second search, Europe PMC: ((TITLE:exercise OR ABSTRACT:exercise) OR (TITLE:"physical activity" OR ABSTRACT:"physical activity") OR (TITLE:prehabilitation OR ABSTRACT:prehabilitation) OR (TITLE:"aerobic training" OR ABSTRACT:"aerobic training") OR (TITLE:"resistance training" OR ABSTRACT:"resistance training")) AND (TITLE:cancer OR ABSTRACT:cancer) AND ((TITLE:"neo-adjuvant" OR ABSTRACT:"neo-adjuvant") OR (TITLE:"(neo)adjuvant" OR ABSTRACT:"(neo)adjuvant") OR (TITLE:neoadjuvant OR ABSTRACT:neoadjuvant) OR (TITLE:"during treatment" OR ABSTRACT:"during treatment") OR (TITLE:"during oncological treatment" OR ABSTRACT:"during oncological treatment")) AND ((TITLE:"cost-effectiveness" OR ABSTRACT:"cost-effectiveness") OR (TITLE:"cost effectiveness" OR ABSTRACT:"cost effectiveness") OR (TITLE:"cost-utility" OR ABSTRACT:"cost-utility") OR (TITLE:"cost utility" OR ABSTRACT:"cost utility") OR (TITLE:"economic evaluation" OR ABSTRACT:"economic evaluation") OR (TITLE:QALY OR ABSTRACT:QALY) OR (TITLE:"quality-adjusted life year" OR ABSTRACT:"quality-adjusted life year")) AND PUB_YEAR:[2000 TO 2026]
Third search, PubMed: (exercise[tiab] OR "physical activity"[tiab] OR "Exercise"[mh] OR "Exercise Therapy"[mh] OR prehabilitation[tiab] OR "aerobic training"[tiab] OR "resistance training"[tiab]) AND ("Breast Neoplasms"[mh] OR "breast cancer"[tiab] OR "breast carcinoma"[tiab]) AND (chemotherapy[tiab] OR "Antineoplastic Agents"[mh]) AND ("cost-effectiveness"[tiab] OR "cost effectiveness"[tiab] OR "cost-utility"[tiab] OR "cost utility"[tiab] OR "economic evaluation"[tiab] OR QALY*[tiab] OR "Cost-Benefit Analysis"[mh]) AND ("2000/01/01"[dp] : "2026/12/31"[dp])
Third search, Europe PMC: ((TITLE:exercise OR ABSTRACT:exercise) OR (TITLE:"physical activity" OR ABSTRACT:"physical activity") OR (TITLE:prehabilitation OR ABSTRACT:prehabilitation) OR (TITLE:"aerobic training" OR ABSTRACT:"aerobic training") OR (TITLE:"resistance training" OR ABSTRACT:"resistance training")) AND ((TITLE:"breast cancer" OR ABSTRACT:"breast cancer") OR (TITLE:"breast neoplasms" OR ABSTRACT:"breast neoplasms") OR (TITLE:"breast carcinoma" OR ABSTRACT:"breast carcinoma")) AND (TITLE:chemotherapy OR ABSTRACT:chemotherapy) AND ((TITLE:"cost-effectiveness" OR ABSTRACT:"cost-effectiveness") OR (TITLE:"cost effectiveness" OR ABSTRACT:"cost effectiveness") OR (TITLE:"cost-utility" OR ABSTRACT:"cost-utility") OR (TITLE:"cost utility" OR ABSTRACT:"cost utility") OR (TITLE:"economic evaluation" OR ABSTRACT:"economic evaluation") OR (TITLE:QALY OR ABSTRACT:QALY) OR (TITLE:"quality-adjusted life year" OR ABSTRACT:"quality-adjusted life year")) AND PUB_YEAR:[2000 TO 2026]

Reconstructed patient-level data

Kaplan-Meier curves of KEYNOTE-522 were digitized from the published figures: event-free survival by arm and pathological complete response (pCR) status at interim analysis 4, with a data cutoff of 23 March 2021 [4], and overall survival by arm at the final analysis of overall survival, with a data cutoff of 22 March 2024 [5]. Patient-level data were reconstructed from the digitized curves and the published numbers at risk with the algorithm of Guyot and colleagues [49]. A reconstruction was accepted when it reproduced each published milestone estimate within 1 percentage point, each published hazard ratio (HR) within 0.03 and each reported number of events within 2%; the reported numbers of events were not given to the algorithm. The proportions with pCR were those of interim analysis 4 (Supplementary Table S2).

Event-free survival after surgery

Event-free survival was modeled from surgery, set at 7 months after randomization, separately for the four combinations of arm and pCR status. The candidate models were six standard parametric

distributions, six Royston-Parmar spline models and three mixture cure models on the relative survival scale, with background mortality from the Japanese life table for women [51]. A model was eligible if at least 95% of the monthly points with at least 20 patients at risk lay within the pointwise 95% confidence interval of the Kaplan-Meier estimate, if its excess hazard peaked within 3 years after surgery and declined thereafter, and if its all-cause hazard at 10 and 20 years was not below background mortality.

Within each pCR stratum, both arms used the same distribution family. Among the families eligible in both arms whose summed Akaike information criterion was within 4 of the lowest, the family whose 5-year extrapolation was closest to the 5-year event-free survival reported at a later data cutoff was selected [43]. This gave a Weibull model for patients with pCR and a Weibull mixture cure model for patients without pCR. Event-free survival at 7 years in each stratum was then fixed at the value of the final analysis [44] (Supplementary Table S2), by maximizing the likelihood subject to this equality.

Events before surgery, first events after surgery and deaths from adverse events

Before surgery, event-free survival in each stratum followed its Kaplan-Meier estimate, and patients with pCR had no events before surgery. Progression that precluded surgery occurred only in patients without pCR, in the proportion reported for each arm [5], with the monthly shape of the Kaplan-Meier estimate before surgery. When the reported proportion exceeded the decline of that estimate, the excess was placed in the first cycle after surgery and taken from the excess hazard, so the total was unchanged.

First events after surgery were divided into locoregional recurrence, including second primary cancers, distant recurrence and death without recurrence, in the proportions of the 2024 report [5]. Locoregional recurrence progressed to distant recurrence at a constant rate [4]. Deaths from adverse events, in the numbers reported for each arm [4], occurred at a constant hazard during the 12 months of treatment and were moved within the fitted event-free survival without changing it.

Survival after recurrence and background mortality

Overall survival curves of patients with early and late relapse in the Epidemiological Strategy and Medical Economics (ESME) cohort [50] were digitized and reconstructed in the same way, and log-logistic distributions were fitted to them. A common shape was used because the shapes of the two groups did not differ significantly. Excess mortality after distant recurrence within 18 months of randomization followed the early-relapse distribution and after later recurrence the late-relapse distribution, in the same way in all arms. Excess mortality after progression that precluded surgery followed the early-relapse distribution.

Survival after locoregional recurrence carried background mortality only. Background mortality came from the 2025 abridged life table for Japanese women [51], from age 49 years. The all-cause hazard was never below background mortality; where a fitted model for patients with pCR fell below it late in follow-up, background mortality applied.

Validation

Overall survival was not used to fit or calibrate any parameter. The model was accepted if three criteria, fixed before the comparison, were met. First, overall survival of each arm at years 1 to 5 lay within the pointwise 95% confidence intervals of the reconstructed Kaplan-Meier estimates, or all ten points lay within Bonferroni-adjusted 99.5% intervals. Second, the 5-year difference in restricted mean survival time between arms lay within the bootstrap 95% interval of the Kaplan-Meier difference. Third, overall and event-free survival at 7 years lay within the 95% confidence intervals of the final analysis [44]. Supplementary Fig. S1 and Supplementary Table S3 show the comparison.

Probabilistic sensitivity analysis

Each of the 1,000 probabilistic samples combined one bootstrap replicate of the survival model with one draw of the other inputs. In each replicate, the reconstructed event-free survival data were resampled within arm together with their pCR status and the models were refitted with the 7-year equality; the 7-year values were drawn from binomial distributions with the stratum sizes of interim analysis 4. The reconstructed ESME data were resampled within group and the log-logistic distributions refitted.

The other inputs were drawn independently from the distributions in Supplementary Table S2: gamma distributions for costs, durations and events per patient, beta distributions for proportions and utilities, and lognormal distributions for hazard ratios, with parameters derived from the 95% ranges. Drug prices and rates based on zero events were fixed, and the duration of the utility decrement for adverse events and the discount rate were not varied. Every sample can be reproduced from its random seed (base seed 20261023, plus the sample number for the bootstrap). The 2.5th and 97.5th percentiles across samples give the 95% uncertainty intervals.

Supplementary Table S1. Consolidated Health Economic Evaluation Reporting Standards 2022 checklist[a]

Title			
Section or topic	Item	Guidance for reporting	Location in the manuscript
Title	1	Identify the study as an economic evaluation and specify the interventions being compared.	Title page (pp. 1 to 2)

Abstract			
Section or topic	Item	Guidance for reporting	Location in the manuscript
Abstract	2	Provide a structured summary that highlights context, key methods, results, and alternative analyses.	Abstract (p. 3)

Introduction			
Section or topic	Item	Guidance for reporting	Location in the manuscript
Background and objectives	3	Give the context for the study, the study question, and its practical relevance for decision making in policy or practice.	Introduction, paragraphs 1 to 4 (pp. 4 to 5)

Methods			
Section or topic	Item	Guidance for reporting	Location in the manuscript
Health economic analysis plan	4	Indicate whether a health economic analysis plan was developed and where available.	Methods, Study design, paragraph 3 (p. 6)
Study population	5	Describe characteristics of the study population (such as age range, demographics, socioeconomic, or clinical characteristics).	Methods, Study design, paragraph 1; Table 1 (pp. 5 to 6)
Setting and location	6	Provide relevant contextual information that may influence findings.	Methods, Study design, paragraphs 1 to 2 (pp. 5 to 6)

Comparators	7	Describe the interventions or strategies being compared and why chosen.	Methods, Study design, paragraph 1; Methods, Exercise effect and costs, paragraph 3 (pp. 5 to 8)
Perspective	8	State the perspective(s) adopted by the study and why chosen.	Methods, Study design, paragraph 2 (p. 6)
Time horizon	9	State the time horizon for the study and why appropriate.	Methods, Study design, paragraph 2 (p. 6)
Discount rate	10	Report the discount rate(s) and reason chosen.	Methods, Study design, paragraph 2 (p. 6)
Selection of outcomes	11	Describe what outcomes were used as the measure(s) of benefit(s) and harm(s).	Methods, Outcomes, paragraphs 1 to 2 (pp. 6 to 7)
Measurement of outcomes	12	Describe how outcomes used to capture benefit(s) and harm(s) were measured.	Methods, Exercise effect and costs, paragraphs 1 to 2; Methods, Survival, costs and utilities, paragraph 1; Supplementary Methods S1 (pp. 7 to 8)
Valuation of outcomes	13	Describe the population and methods used to measure and value outcomes.	Methods, Survival, costs and utilities, paragraph 2 (pp. 8 to 9)
Measurement and valuation of resources and costs	14	Describe how costs were valued.	Methods, Exercise effect and costs, paragraph 3; Methods, Survival, costs and utilities, paragraph 2; Table 1; Supplementary Table S2 (pp. 7 to 9)
Currency, price date, and conversion	15	Report the dates of the estimated resource quantities and unit costs, plus the currency and year of conversion.	Methods, Exercise effect and costs, paragraph 3; Methods, Survival, costs and utilities, paragraph 2 (pp. 7 to 9)
Rationale and description of model	16	If modelling is used, describe in detail and why used. Report if the model is publicly available and where it can be accessed.	Methods, Study design, paragraph 1; Methods, Statistical analysis, paragraph 1; Fig. 1; Supplementary Methods S1; Statements and Declarations, Data availability (pp.

			5 to 29)
Analytics and assumptions	17	Describe any methods for analysing or statistically transforming data, any extrapolation methods, and approaches for validating any model used.	Methods, Exercise effect and costs, paragraphs 1 to 2; Methods, Survival, costs and utilities, paragraph 1; Methods, Statistical analysis, stages (1) to (3); Supplementary Methods S1 (pp. 7 to 9)
Characterising heterogeneity	18	Describe any methods used for estimating how the results of the study vary for subgroups.	No subgroup analyses beyond exercise added to chemotherapy alone and to chemotherapy plus pembrolizumab; Methods, Study design, paragraph 1 (pp. 5 to 6)
Characterising distributional effects	19	Describe how impacts are distributed across different individuals or adjustments made to reflect priority populations.	Not performed
Characterising uncertainty	20	Describe methods to characterise any sources of uncertainty in the analysis.	Methods, Statistical analysis, stages (4) to (6); Supplementary Methods S1; Discussion, Limitations, paragraph 2 (pp. 9 to 15)
Approach to engagement with patients and others affected by the study	21	Describe any approaches to engage patients or service recipients, the general public, communities, or stakeholders (such as clinicians or payers) in the design of the study.	Not performed

Results			
Section or topic	Item	Guidance for reporting	Location in the manuscript
Study parameters	22	Report all analytic inputs (such as values, ranges, references) including uncertainty or distributional assumptions.	Table 1; Supplementary Table S2

Summary of main results	23	Report the mean values for the main categories of costs and outcomes of interest and summarise them in the most appropriate overall measure.	Results, Four strategies compared, paragraph 1; Results, Exercise compared with pembrolizumab, paragraph 1; Results, Maximum affordable cost of exercise, paragraphs 1 to 3; Results, Duration and metastatic treatment costs, paragraphs 1 to 3; Fig. 2; Tables 2 and 3; Supplementary Tables S4 to S9 and S12 (pp. 10 to 12)
Effect of uncertainty	24	Describe how uncertainty about analytic judgments, inputs, or projections affect findings. Report the effect of choice of discount rate and time horizon, if applicable.	Results, Duration and metastatic treatment costs, paragraphs 1 to 2; Results, Sensitivity and scenario analyses, paragraph 1; Fig. 3; Table 3; Supplementary Tables S5, S8, S10 and S11; Supplementary Fig. S2 (pp. 11 to 12)
Effect of engagement with patients and others affected by the study	25	Report on any difference patient/service recipient, general public, community, or stakeholder involvement made to the approach or findings of the study	Not applicable (item 21)

Discussion			
Section or topic	Item	Guidance for reporting	Location in the manuscript
Study findings, limitations, generalisability, and current knowledge	26	Report key findings, limitations, ethical or equity considerations not captured, and how these could affect patients, policy, or practice.	Discussion, paragraphs 1 to 7; Discussion, Implications; Discussion, Limitations (pp. 12 to 15)

Other relevant information			
Section or topic	Item	Guidance for reporting	Location in the manuscript
Source of funding	27	Describe how the study was funded	Statements and Declarations,

		and any role of the funder in the identification, design, conduct, and reporting of the analysis	Funding (p. 28)
Conflicts of interest	28	Report authors conflicts of interest according to journal or International Committee of Medical Journal Editors requirements.	Statements and Declarations, Competing interests (pp. 28 to 29)

[a]Checklist and guidance as published [45]. Locations give the section, the paragraph within the section or subsection (or the numbered stage of the statistical analysis) and the page of the manuscript file; tables and figures are named where they report the item. CHEERS, Consolidated Health Economic Evaluation Reporting Standards.

Supplementary Table S2. All model inputs[a]

A. Survival and model structure				
Input	Arms	Base case	Bootstrap 2.5th to 97.5th percentile	Source
Starting age, years	All arms	49	—	KEYNOTE-522 population [3]
Pathological complete response (pCR)	With pembrolizumab	494/784 (63.0%)	—	KEYNOTE-522 (interim analysis 4) [4]
Pathological complete response (pCR)	Without pembrolizumab	217/390 (55.6%)	—	KEYNOTE-522 (interim analysis 4) [4]
Event-free survival at 7 years, pCR (fixed in fitting)	With pembrolizumab	90.4%	87.7 to 92.9%	KEYNOTE-522 final analysis [44]
Event-free survival at 7 years, no pCR (fixed in fitting)	With pembrolizumab	57.6%	52.1 to 63.4%	KEYNOTE-522 final analysis [44]
Event-free survival at 7 years, pCR (fixed in fitting)	Without pembrolizumab	85.9%	81.1 to 90.3%	KEYNOTE-522 final analysis [44]
Event-free survival at 7 years, no pCR (fixed in fitting)	Without pembrolizumab	49.7%	41.6 to 57.8%	KEYNOTE-522 final analysis [44]
Event-free survival after surgery, pCR: Weibull, shape	With pembrolizumab	0.6685	0.5106 to 0.9761	Fitted to reconstructed KEYNOTE-522 data (interim analysis 4) [4,49]
Event-free survival after surgery, pCR: Weibull, scale (years)	With pembrolizumab	198.2	59.79 to 803.0	Fitted to reconstructed KEYNOTE-522 data (interim analysis 4) [4,49]
Event-free survival after surgery, no pCR: Weibull cure model, cure fraction	With pembrolizumab	0.5961	0.3655 to 0.6769	Fitted to reconstructed KEYNOTE-522 data (interim analysis 4) [4,49]
Event-free survival after surgery, no pCR: Weibull cure model, shape	With pembrolizumab	0.8468	0.6664 to 1.145	Fitted to reconstructed KEYNOTE-522 data (interim analysis 4) [4,49]
Event-free survival after	With pembrolizumab	1.683	0.9405 to 5.291	Fitted to reconstructed

surgery, no pCR: Weibull cure model, scale (years)				KEYNOTE-522 data (interim analysis 4) [4,49]
Event-free survival after surgery, pCR: Weibull, shape	Without pembrolizumab	0.7593	0.5748 to 1.164	Fitted to reconstructed KEYNOTE-522 data (interim analysis 4) [4,49]
Event-free survival after surgery, pCR: Weibull, scale (years)	Without pembrolizumab	76.71	28.14 to 284.7	Fitted to reconstructed KEYNOTE-522 data (interim analysis 4) [4,49]
Event-free survival after surgery, no pCR: Weibull cure model, cure fraction	Without pembrolizumab	0.5427	0.4249 to 0.6334	Fitted to reconstructed KEYNOTE-522 data (interim analysis 4) [4,49]
Event-free survival after surgery, no pCR: Weibull cure model, shape	Without pembrolizumab	1.216	0.9402 to 1.627	Fitted to reconstructed KEYNOTE-522 data (interim analysis 4) [4,49]
Event-free survival after surgery, no pCR: Weibull cure model, scale (years)	Without pembrolizumab	1.424	1.038 to 2.451	Fitted to reconstructed KEYNOTE-522 data (interim analysis 4) [4,49]
Progression precluding surgery, proportion of patients without pCR	With pembrolizumab	4.83%	—	KEYNOTE-522 (2024 report) [5]
Progression precluding surgery, proportion of patients without pCR	Without pembrolizumab	8.67%	—	KEYNOTE-522 (2024 report) [5]
Composition of first events after surgery	With pembrolizumab	locoregional recurrence 37.1%; distant recurrence 58.3%; death without recurrence 4.6%	—	KEYNOTE-522 (2024 report) [5]
Composition of first events after surgery	Without pembrolizumab	locoregional recurrence 31.5%; distant recurrence 58.8%; death without recurrence 9.8%	—	KEYNOTE-522 (2024 report) [5]
Deaths from adverse events during treatment	With pembrolizumab	7/783	—	KEYNOTE-522 (interim analysis 4) [4]
Deaths from adverse events	Without pembrolizumab	1/389	—	KEYNOTE-522 (interim

during treatment				analysis 4) [4]
Locoregional to distant recurrence, rate per year	All arms	0.340 (22 of 45 patients; 1.98 years of follow-up)	—	KEYNOTE-522 (2022 report) [4]
Excess mortality after distant recurrence within 18 months of randomization: log-logistic median, months	All arms	9.73	9.10 to 10.43	ESME cohort, reconstructed overall survival [49,50]
Excess mortality after later distant recurrence: log-logistic median, months	All arms	16.93	15.88 to 18.01	ESME cohort, reconstructed overall survival [49,50]
Excess mortality after distant recurrence or progression during neoadjuvant therapy: common log-logistic shape	All arms	1.660	1.590 to 1.736	ESME cohort, reconstructed overall survival [49,50]
Excess mortality after progression during neoadjuvant therapy	All arms	Same as distant recurrence within the first 18 months	As above	ESME cohort [50]
Background mortality	All arms	Japanese women	—	Abridged life table (2025) [51]
Scenario: Higher mortality after early recurrence with pembrolizumab, multiplier	With pembrolizumab	1.307	—	Italian real-world cohort after neoadjuvant pembrolizumab [66]
Scenario: progression during neoadjuvant therapy with longer survival, log-logistic median, months	All arms	26	—	Single-center cohort with locoregional progression during neoadjuvant therapy [67]

B. Costs, adverse events, utilities, exercise and post-operative treatment							
Code	Input	Arms	Unit	Base case	Range	Distribution	Source
A01	Paclitaxel plus carboplatin (12 weeks), drug cost	All arms	JPY per 12 weeks	113,736	85,302 to 113,736 [drug price, lower only]	Fixed	National drug price list [8]

A02	Doxorubicin or epirubicin plus cyclophosphamide (4 cycles), drug cost	All arms	JPY per 4 cycles	33,496	25,122 to 33,496 [drug price, lower only]	Fixed	National drug price list [8]
A03	Pembrolizumab, drug cost per dose	With pembrolizumab	JPY per administration	398,924	299,193 to 398,924 [drug price, lower only]	Fixed	National drug price list [8]
A04	Outpatient drug administration, per visit (fees and tests included)	All arms	JPY per administration	15,565	10,768 to 18,160	Gamma (shape 68.12, scale 228.5)	Medical fee schedule [48]
A05	Supportive care (G-CSF, antiemetics), per cycle	All arms	JPY per cycle	36,558	5,820 to 67,296 [mirrored (assumed)]	Gamma (shape 5.434, scale 6,728)	National drug price list; antiemetic guideline [8,52]
A06	Surgery (mix of procedures)	All arms	JPY per event	1,000,000	641,441 to 1,250,000	Gamma (shape 41.49, scale 24,102)	Published cost estimates of Kasugai Municipal Hospital and Chuden Hospital; medical fee schedule code K476; KEYNOTE-522 surgical outcomes [48,53–55]
A07	Postoperative radiotherapy, proportion treated	All arms	proportion	0.63230	0.50600 to 0.90000	Beta (alpha 13.92, beta 8.094)	KEYNOTE-522 surgical outcomes [55]

A08	Postoperative radiotherapy, cost per course	All arms	JPY per course	445,300	405,300 to 485,300 [mirrored (assumed)]	Gamma (shape 476.1, scale 935.3)	Medical fee schedule [48]
B01	Immune-related AE management cost: hypothyroidism	All arms	JPY per event	38,586	19,293 to 236,000	Gamma (shape 0.4872, scale 79,206)	Published Japanese estimate [76]
B02	Immune-related AE management cost: hyperthyroidism	All arms	JPY per event	38,856	19,428 to 236,000	Gamma (shape 0.4946, scale 78,558)	Published Japanese estimate [76]
B03	Immune-related AE management cost: adrenal insufficiency	All arms	JPY per event	236,000	118,000 to 472,000	Gamma (shape 6.829, scale 34,557)	Published Japanese estimate, used as a proxy [56]
B04	Immune-related AE management cost: hypophysitis	All arms	JPY per event	236,000	118,000 to 472,000	Gamma (shape 6.829, scale 34,557)	Published Japanese estimate, used as a proxy [56]
B05	Immune-related AE management cost: pneumonitis	All arms	JPY per event	63,257	31,628 to 236,000	Gamma (shape 1.472, scale 42,971)	Published Japanese estimate [76]
B06	Immune-related AE management cost: colitis	All arms	JPY per event	40,740	20,370 to 236,000	Gamma (shape 0.5485, scale 74,275)	Published Japanese estimate [76]
B07	Immune-related AE management cost: hepatitis	All arms	JPY per event	40,740	20,370 to 236,000	Gamma (shape 0.5485, scale 74,275)	Published Japanese estimate [76]
B08	Immune-related AE management cost: severe skin reactions	All arms	JPY per event	236,000	38,057 to 472,000	Gamma (shape 4.545, scale 51,928)	Published Japanese estimates [56,76]
B09	Immune-related	All arms	JPY per event	236,000	118,000 to	Gamma (shape	Published

	AE management cost: type 1 diabetes				472,000	6.829, scale 34,557)	Japanese estimate, used as a proxy [56]
B10	Immune-related AE management cost: other serious events	All arms	JPY per event	236,000	118,000 to 472,000	Gamma (shape 6.829, scale 34,557)	Published Japanese estimate, used as a proxy [56]
B11	Febrile neutropenia management cost per event	All arms	JPY per event	236,000	189,000 to 283,000 [mirrored (assumed)]	Gamma (shape 96.86, scale 2,437)	Published Japanese estimate, used as a proxy [56]
B21P	Incidence: hypothyroidism (immune-related)	With pembrolizumab	proportion	0.15070	0.12636 to 0.17771	Beta (alpha 112.2, beta 632.6)	KEYNOTE-522 (interim analysis 4) [4]
B21C	Incidence: hypothyroidism (immune-related)	Without pembrolizumab	proportion	0.05656	0.03578 to 0.08438	Beta (alpha 19.58, beta 326.6)	KEYNOTE-522 (interim analysis 4) [4]
B22P	Incidence: hyperthyroidism (immune-related)	With pembrolizumab	proportion	0.05236	0.03783 to 0.07037	Beta (alpha 37.65, beta 681.4)	KEYNOTE-522 (interim analysis 4) [4]
B22C	Incidence: hyperthyroidism (immune-related)	Without pembrolizumab	proportion	0.01799	0.00726 to 0.03672	Beta (alpha 5.609, beta 306.2)	KEYNOTE-522 (interim analysis 4) [4]
B23P	Incidence: adrenal insufficiency (immune-related)	With pembrolizumab	proportion	0.02554	0.01567 to 0.03917	Beta (alpha 17.66, beta 673.8)	KEYNOTE-522 (interim analysis 4) [4]
B23C	Incidence: adrenal insufficiency (immune-related)	Without pembrolizumab	proportion	0.00000	0.00000 to 0.00944 [zero events, fixed in PSA]	Fixed	KEYNOTE-522 (interim analysis 4) [4]
B24P	Incidence: hypophysitis (immune-related)	With pembrolizumab	proportion	0.01916	0.01076 to 0.03140	Beta (alpha 12.97, beta 663.9)	KEYNOTE-522 (interim analysis 4) [4]
B24C	Incidence:	Without	proportion	0.00257	0.00007 to	Beta (alpha	KEYNOTE-522

	hypophysitis (immune-related)	pembrolizumab			0.01424	0.5016, beta 194.7)	(interim analysis 4) [4]
B25P	Incidence: pneumonitis (immune-related)	With pembrolizumab	proportion	0.00894	0.00360 to 0.01833	Beta (alpha 5.601, beta 620.9)	KEYNOTE-522 (interim analysis 4) [4]
B25C	Incidence: pneumonitis (immune-related)	Without pembrolizumab	proportion	0.00514	0.00062 to 0.01845	Beta (alpha 1.265, beta 244.9)	KEYNOTE-522 (interim analysis 4) [4]
B26P	Incidence: colitis (immune-related)	With pembrolizumab	proportion	0.00766	0.00282 to 0.01660	Beta (alpha 4.704, beta 609.4)	European Medicines Agency assessment report [61]
B26C	Incidence: colitis (immune-related)	Without pembrolizumab	proportion	0.00000	0.00000 to 0.00944 [zero events, fixed in PSA]	Fixed	European Medicines Agency assessment report [61]
B27P	Incidence: hepatitis (immune-related)	With pembrolizumab	proportion	0.01149	0.00527 to 0.02171	Beta (alpha 7.408, beta 637.3)	European Medicines Agency assessment report [61]
B27C	Incidence: hepatitis (immune-related)	Without pembrolizumab	proportion	0.00000	0.00000 to 0.00944 [zero events, fixed in PSA]	Fixed	European Medicines Agency assessment report [61]
B28P	Incidence: severe skin reactions (immune-related)	With pembrolizumab	proportion	0.04725	0.03349 to 0.06455	Beta (alpha 33.83, beta 682.2)	KEYNOTE-522 (interim analysis 4) [4]
B28C	Incidence: severe skin reactions (immune-related)	Without pembrolizumab	proportion	0.00257	0.00007 to 0.01424	Beta (alpha 0.5016, beta 194.7)	KEYNOTE-522 (interim analysis 4) [4]

B29P	Incidence: type 1 diabetes (immune-related)	With pembrolizumab	proportion	0.00511	0.00139 to 0.01303	Beta (alpha 2.941, beta 572.6)	European Medicines Agency assessment report [61]
B29C	Incidence: type 1 diabetes (immune-related)	Without pembrolizumab	proportion	0.00000	0.00000 to 0.00944 [zero events, fixed in PSA]	Fixed	European Medicines Agency assessment report [61]
B30P	Incidence: other serious immune-related AEs	With pembrolizumab	proportion	0.02043	0.01172 to 0.03297	Beta (alpha 13.89, beta 666.1)	KEYNOTE-522 (interim analysis 4); European Medicines Agency assessment report [4,61]
B30C	Incidence: other serious immune-related AEs	Without pembrolizumab	proportion	0.00257	0.00007 to 0.01424	Beta (alpha 0.5016, beta 194.7)	KEYNOTE-522 (interim analysis 4); European Medicines Agency assessment report [4,61]
B31P	Incidence: febrile neutropenia	With pembrolizumab	proportion	0.18391	0.15737 to 0.21285	Beta (alpha 137.6, beta 610.6)	KEYNOTE-522 supplementary appendix (interim analysis 4) [4]
B31C	Incidence: febrile neutropenia	Without pembrolizumab	proportion	0.16195	0.12675 to 0.20240	Beta (alpha 58.85, beta 304.6)	KEYNOTE-522 supplementary appendix

							(interim analysis 4) [4]
B32P	Incidence: any grade 3 or higher AE (scenario)	With pembrolizumab	proportion	0.82375	0.79522 to 0.84982	Beta (alpha 615.6, beta 131.7)	KEYNOTE-522 (interim analysis 4) [4]
B32C	Incidence: any grade 3 or higher AE (scenario)	Without pembrolizumab	proportion	0.78663	0.74253 to 0.82631	Beta (alpha 288.2, beta 78.19)	KEYNOTE-522 (interim analysis 4) [4]
B33P	Grade 3 or higher AEs of six types, events per patient	With pembrolizumab	events per patient	1.064	0.993 to 1.139	Gamma (shape 818.2, scale 0.001300)	KEYNOTE-522 supplementary appendix (interim analysis 4) [4]
B33C	Grade 3 or higher AEs of six types, events per patient	Without pembrolizumab	events per patient	0.982	0.886 to 1.086	Gamma (shape 371.9, scale 0.002640)	KEYNOTE-522 supplementary appendix (interim analysis 4) [4]
C01	Follow-up cost per year, years 1 to 2	All arms	JPY per year	18,270	11,660 to 24,880 [mirrored (assumed)]	Gamma (shape 29.35, scale 622.5)	Medical fee schedule [48]
C02	Follow-up cost per year, years 3 to 5	All arms	JPY per year	16,730	10,120 to 23,340 [mirrored (assumed)]	Gamma (shape 24.61, scale 679.8)	Medical fee schedule [48]
C03	Follow-up cost per year, year 6 onward	All arms	JPY per year	15,960	9,350 to 22,570 [mirrored (assumed)]	Gamma (shape 22.40, scale 712.6)	Medical fee schedule [48]
D01	Locoregional recurrence, initial treatment cost	All arms	JPY per event	1,222,650	916,988 to 1,528,313 [±25% (assumed)]	Gamma (shape 61.46, scale 19,892)	Derived from inputs A06 and A08
D02	Locoregional	All arms	JPY per month	3,045	2,284 to 3,806	Gamma (shape	Derived from

	recurrence, monthly cost thereafter				[±25% (assumed)]	61.46, scale 49.54)	input C01
D03	First-line treatment of distant recurrence, monthly cost	Without pembrolizumab	JPY per month	1,215,933	915,446 to 1,416,257	Gamma (shape 90.58, scale 13,424)	A breast medical oncologist at a Japanese hospital (personal communication, October 2026); trials; national drug price list [8,72–75]
D03P	First-line treatment of distant recurrence, monthly cost	With pembrolizumab	JPY per month	1,458,318	973,662 to 1,781,422	Gamma (shape 50.08, scale 29,118)	A breast medical oncologist at a Japanese hospital (personal communication, October 2026); trials; national drug price list [8,72–75]
D04	First-line treatment of distant recurrence, mean duration	Without pembrolizumab	months	10.37	7.77 to 12.96 [±25% (assumed)]	Gamma (shape 61.46, scale 0.1686)	A breast medical oncologist at a Japanese hospital (personal communication, October 2026); trials; national

							drug price list [8,72–75]
D04P	First-line treatment of distant recurrence, mean duration	With pembrolizumab	months	10.84	8.13 to 13.55 [±25% (assumed)]	Gamma (shape 61.46, scale 0.1764)	A breast medical oncologist at a Japanese hospital (personal communication, October 2026); trials; national drug price list [8,72–75]
D05	Second-line and later treatment, monthly cost	All arms	JPY per month	1,412,953	1,059,715 to 1,766,192 [±25% (assumed)]	Gamma (shape 61.46, scale 22,989)	A breast medical oncologist at a Japanese hospital (personal communication, October 2026); national drug price list [8]
D06	End-of-life care, last 3 months	All arms	JPY per 3 months	1,120,001	956,901 to 1,283,100 [mirrored (assumed)]	Gamma (shape 181.1, scale 6,183)	Published Japanese estimates [76,77]
E01	CHALLENGE-based supervised program, total over 6 months	With exercise	JPY per 6 months	239,508	95,803 to 322,320	Gamma (shape 17.18, scale 13,942)	CHALLENGE cost-utility analysis; purchasing power parity and exchange rate [46,47,78]

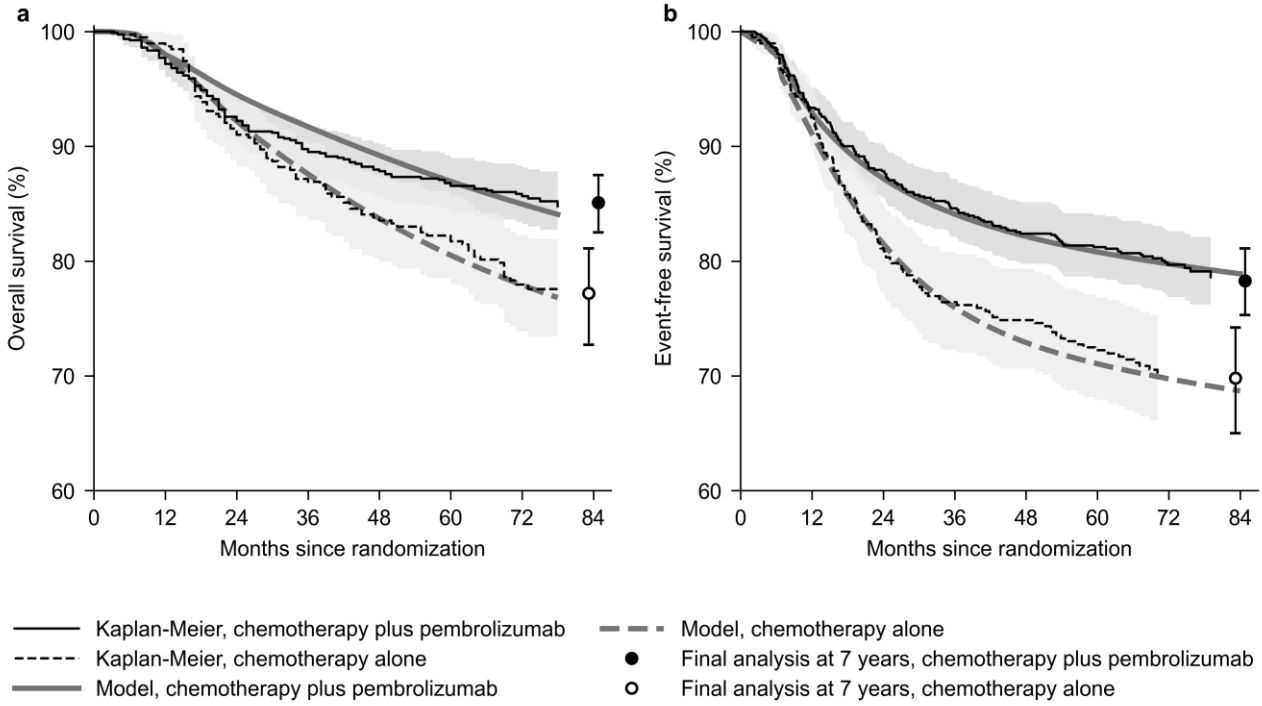
E02	Fee-schedule-based program, total over 6 months	With exercise	JPY per 6 months	192,400	96,200 to 232,440	Gamma (shape 30.64, scale 6,278)	Medical fee schedule [48]
E03	Remote low-cost program, total over 6 months	With exercise	JPY per 6 months	11,260	8,445 to 14,075 [±25% (assumed)]	Gamma (shape 61.46, scale 183.2)	Medical fee schedule [48]
E04	Exercise continued after surgery, monthly cost (scenario)	With exercise	JPY per month	3,700	1,260 to 7,400	Gamma (shape 5.580, scale 663.1)	Medical fee schedule [48]
E05	Exercise continued after surgery, duration (scenario)	With exercise	months	30.00	6.00 to 30.00	Gamma (shape 24.01, scale 1.250)	CHALLENGE trial [9]
E06	Musculoskeletal AEs, any grade, with exercise	With exercise	proportion	0.18458	0.14895 to 0.22464	Beta (alpha 74.33, beta 328.4)	CHALLENGE trial [9]
E07	Musculoskeletal AEs, any grade, without exercise	Without exercise	proportion	0.11497	0.08731 to 0.14767	Beta (alpha 49.22, beta 378.9)	CHALLENGE trial [9]
E08	Musculoskeletal AE management cost per event	All arms	JPY per event	8,810	6,540 to 11,080 [mirrored (assumed)]	Gamma (shape 57.86, scale 152.3)	Medical fee schedule [48]
E09	Musculoskeletal AEs, grade 3 or higher, with exercise	With exercise	proportion	0.00701	0.00145 to 0.02035	Beta (alpha 2.092, beta 296.3)	CHALLENGE trial [9]
E10	Musculoskeletal AEs, grade 3 or higher, without exercise	Without exercise	proportion	0.00217	0.00005 to 0.01203	Beta (alpha 0.5009, beta 230.3)	CHALLENGE trial [9]
F01	Capecitabine use among gBRCA-	Without pembrolizumab	proportion	0.85000	0.80000 to 0.90000	Beta (alpha 165.7, beta	A breast medical

	negative patients without pCR					29.24)	oncologist at a Japanese hospital (personal communication, October 2026)
F01P	Capecitabine use among gBRCA-negative patients without pCR	With pembrolizumab	proportion	0.50000	0.05000 to 0.90000	Beta (alpha 2.158, beta 2.158)	Two breast medical oncologists at Japanese hospitals (personal communication, October 2026); real-world and expert-consensus studies [57,58]
F02	Capecitabine cost, whole course	All arms	JPY per patient	121,402	91,051 to 203,251	Gamma (shape 17.99, scale 6,748)	National drug price list; CREATE-X [8,69]
F03	gBRCA-positive proportion among patients with TNBC	All arms	proportion	0.20000	0.15000 to 0.25000 [±25% (assumed)]	Beta (alpha 48.97, beta 195.9)	Japanese single-institution study [59]
F04	Olaparib use among gBRCA-positive patients without pCR	Without pembrolizumab	proportion	0.90000	0.70000 to 1.00000	Beta (alpha 12.93, beta 1.437)	A breast medical oncologist at a Japanese hospital (personal communication, October 2026)

F04P	Olaparib use among gBRCA-positive patients without pCR	With pembrolizumab	proportion	0.90000	0.50000 to 1.00000	Beta (alpha 4.079, beta 0.4532)	Two breast medical oncologists at Japanese hospitals (personal communication, October 2026)
F05	Olaparib cost, whole course	All arms	JPY per patient	7,079,040	3,539,520 to 7,079,040 [drug price, lower only]	Fixed	National drug price list; olaparib package insert [8,60]
F06	BRCA test cost	All arms	JPY per event	203,000	193,000 to 213,000 [mirrored (assumed)]	Gamma (shape 1,583, scale 128.2)	Medical fee schedule [48]
F07	Capecitabine HR for invasive disease or recurrence	All arms	HR	0.58	0.39 to 0.87	Lognormal (meanlog −0.5447, sdlog 0.2047)	CREATE-X [69]
F08	Olaparib HR for invasive disease	All arms	HR	0.65	0.53 to 0.78	Lognormal (meanlog −0.4308, sdlog 0.09858)	OlympiA, updated analysis [71]
U01	Utility, event-free during treatment	All arms	utility	0.889	0.881 to 0.897	Beta (alpha 5,538, beta 691.5)	KEYNOTE-522 EQ-5D-5L, German value set [30]
U02	Utility, event-free after treatment	All arms	utility	0.887	0.877 to 0.897	Beta (alpha 3,555, beta 452.9)	KEYNOTE-522 EQ-5D-5L, German value set [30]
U03	Utility,	All arms	utility	0.830	0.787 to 0.873	Beta (alpha	KEYNOTE-522

	locoregional recurrence					241.4, beta 49.43)	EQ-5D-5L, German value set [30]
U04	Utility, distant recurrence or progression during neoadjuvant therapy	All arms	utility	0.735	0.700 to 0.770	Beta (alpha 440.6, beta 158.9)	KEYNOTE-522 EQ-5D-5L, German value set [30]
U05	Utility decrement per grade 3 or higher AE (12.5 weeks)	All arms	utility decrement	0.022	0.000 to 0.059	Beta (alpha 2.067, beta 91.91)	KEYNOTE-522 EQ-5D-5L, German value set [30]

[a]Modeled cohort: women aged 49 years starting neoadjuvant chemotherapy for stage II or III triple-negative breast cancer in Japan (Table 1). Costs are in JPY at fiscal year 2026 prices except the CHALLENGE-based supervised program (converted from 2024 Canadian dollars; Table 1). Panel A: base-case values of the survival model and the 2.5th to 97.5th percentiles across 1,000 bootstrap replicates (dash, fixed in all replicates). Panel B: ranges are those of the one-way sensitivity analysis and, for distributions other than fixed, the 95% range of the probabilistic analysis; brackets mark ranges set by rule: [±25% (assumed)], ±25% around the base value; [mirrored (assumed)], the missing bound mirrored; [drug price, lower only], lower bound 75% of the list price and the value fixed in the probabilistic analysis; [zero events, fixed in PSA], upper bound from zero events. Inputs with suffix P or C apply to arms with or without pembrolizumab. AE, adverse event; BRCA, breast cancer gene; ESME, Epidemiological Strategy and Medical Economics database; G-CSF, granulocyte colony-stimulating factor; HR, hazard ratio; JPY, Japanese yen; pCR, pathological complete response; PSA, probabilistic sensitivity analysis; TNBC, triple-negative breast cancer.



Supplementary Fig. S1. Overall survival and event-free survival in KEYNOTE-522: reconstructed Kaplan-Meier curves, the model and the final analysis[a]

(a) Overall survival. (b) Event-free survival. Thin step lines: Kaplan-Meier estimates from patient-level data reconstructed from the published curves of the 2024 report [5,49] (solid, chemotherapy plus pembrolizumab; dashed, chemotherapy alone), with their 95% CIs shaded. Thick grey lines: the model (solid and dashed as before). Points with bars: 7-year estimates of the final analysis [44] with 95% CIs (filled, chemotherapy plus pembrolizumab; open, chemotherapy alone). Overall survival was not used to fit the model. Values are given in Supplementary Table S3.

[a]CI, confidence interval.

Supplementary Table S3. Comparison of the survival model with KEYNOTE-522 overall survival and event-free survival[a]

A. Overall survival, %, at years 1 to 5				
Year	Kaplan-Meier, chemotherapy plus pembrolizumab [95% CI]	Model, chemotherapy plus pembrolizumab	Kaplan-Meier, chemotherapy alone [95% CI]	Model, chemotherapy alone
1	97.2 [96.0, 98.4]	98.0	98.7 [97.6, 99.8]	97.9
2	92.2 [90.3, 94.1]	94.5[b]	91.0 [88.2, 93.9]	92.2
3	89.5 [87.4, 91.7]	91.7[b]	86.9 [83.6, 90.3]	87.6
4	87.8 [85.6, 90.2]	89.2	83.5 [79.9, 87.3]	83.7
5	86.6 [84.2, 89.0]	86.9	81.7 [77.9, 85.6]	80.5

B. Restricted mean survival time		
Quantity	Kaplan-Meier [bootstrap 95% CI]	Model
Difference in 5-year restricted mean survival time, years	0.0985 [−0.0381, 0.2421]	0.1524

C. Final analysis at 7 years			
Quantity at 7 years	Arm	Final analysis [95% CI]	Model
Overall survival, %	Chemotherapy plus pembrolizumab	85.1 [82.5, 87.5]	83.2
Overall survival, %	Chemotherapy alone	77.2 [72.7, 81.1]	75.8
Event-free survival, %	Chemotherapy plus pembrolizumab	78.3 [75.3, 81.1]	78.3
Event-free survival, %	Chemotherapy alone	69.8 [65.0, 74.2]	69.8
Overall survival with pCR, %	Chemotherapy plus pembrolizumab	94.5[c]	93.2
Overall survival with pCR, %	Chemotherapy alone	91.1[c]	90.2
Overall survival without pCR, %	Chemotherapy plus pembrolizumab	69.0[c]	66.3
Overall survival without pCR, %	Chemotherapy alone	59.8[c]	57.7

D. Model extrapolation

Quantity	Arm	10 years [bootstrap 2.5th to 97.5th percentile]	20 years [bootstrap 2.5th to 97.5th percentile]
Overall survival, %	Chemotherapy plus pembrolizumab	79.3 [76.5, 82.0]	71.6 [65.2, 76.0]
Overall survival, %	Chemotherapy alone	71.1 [66.6, 75.3]	61.7 [52.3, 68.0]
Event-free survival, %	Chemotherapy plus pembrolizumab	76.0 [72.0, 79.7]	70.7 [62.6, 75.5]
Event-free survival, %	Chemotherapy alone	67.3 [61.1, 72.1]	60.1 [48.4, 67.1]

[a]Kaplan-Meier estimates from patient-level data reconstructed from the published KEYNOTE-522 curves (overall survival, 2024 report) [5,49]; final analysis [44]: data cutoff 14 October 2025. Overall survival was not used to fit the model; event-free survival at 7 years in each pCR stratum was fixed at the final-analysis values during fitting (Table 1); the model was accepted if the 10 points of panel A lay within their pointwise 95% CIs or within Bonferroni-adjusted 99.5% CIs, the difference in panel B lay within its bootstrap 95% CI, and the four 7-year values with a CI in panel C lay within their 95% CIs. Panel D: 2.5th and 97.5th percentiles across 1,000 bootstrap replicates. CI, confidence interval; pCR, pathological complete response.
[b]Outside the pointwise 95% CI and within the Bonferroni-adjusted 99.5% CI.
[c]No CI reported; shown for comparison only and not part of the acceptance rule.

Supplementary Table S4. Incremental costs by category, QALYs and cost-effectiveness of pembrolizumab added to neoadjuvant chemotherapy, in the base case and under scenarios[a]

Quantity	Base case	Higher mortality after early recurrence with pembrolizumab	Japanese post-operative treatment[b]	Metastatic treatment costs halved[c]	Eribulin-based later-line treatment[c]
A. Incremental costs by category					
Pembrolizumab	6,531	6,531	—	6,531	6,531
Drug administration	132	132	—	132	132
Immune-related adverse events	32	32	—	32	32
Febrile neutropenia	5	5	—	5	5
Follow-up	42	42	—	42	42
Locoregional recurrence	−24	−24	—	−24	−24
Distant recurrence, first-line treatment	−299	−338	—	−150	−299
Distant recurrence, second-line and later treatment	−1,911	−2,197	—	−955	−332
End-of-life care	−101	−101	—	−101	−101
Other (chemotherapy drugs, supportive care, surgery, radiotherapy)	19	19	—	19	19
Total	4,426 [−330, 8,901]	4,102 [−744, 8,558]	4,990	5,531 [3,093, 7,822]	6,005 [3,759, 8,229]
B. Health outcomes and cost-effectiveness					
Incremental QALYs	2.156 [0.245, 4.228]	2.137 [0.230, 4.207]	1.639	2.156 [0.245, 4.228]	2.156 [0.245, 4.228]
ICER, JPY 1,000 per QALY	2,053	1,920	3,044	2,565	2,785
Probability cost-effective at JPY 5 million per QALY, %	83.1	84.4	—	82.1	80.3
Probability cost-effective at JPY 7.5	90.8	91.1	—	91.3	90.5

million per QALY, %					

[a]Modeled cohort: women aged 49 years starting neoadjuvant chemotherapy for stage II or III triple-negative breast cancer in Japan (Table 1). Chemotherapy plus pembrolizumab compared with chemotherapy alone. Cells show deterministic estimates; where probabilistic analysis was run, the 95% UI [2.5th and 97.5th percentiles of 1,000 probabilistic sensitivity analysis samples] follows in brackets. Probability cost-effective: share of samples with a positive net monetary benefit. Costs in JPY 1,000 at fiscal year 2026 prices, discounted at 2% per year over a lifetime horizon; categories may not add to the total because of rounding. Scenarios are defined in Table 1. A dash means not computed. ICER, incremental cost-effectiveness ratio; JPY, Japanese yen; QALY, quality-adjusted life year; UI, uncertainty interval.

[b]Probabilistic analysis and the cost decomposition were not run for this scenario.

[c]Category values equal the base-case values with the distant-recurrence treatment costs multiplied by the scenario factor in Table 1 (survival is unchanged); they add to the modeled total.

Supplementary Table S5. Maximum affordable cost of an exercise program per patient by the recurrence hazard ratio with exercise[a]

A. Threshold JPY 5 million per QALY						
Recurrence HR	Exercise added to chemotherapy plus pembrolizumab, base case	Exercise added to chemotherapy plus pembrolizumab, higher mortality after early recurrence with pembrolizumab	Exercise added to chemotherapy alone	Probability above the fee-schedule-based program cost, %: exercise added to chemotherapy plus pembrolizumab, base case	Same, higher mortality after early recurrence with pembrolizumab	Same, exercise added to chemotherapy alone
0.700	5,596 [4,545, 6,256]	5,524 [4,491, 6,164]	7,107 [5,923, 7,869]	100.0	100.0	100.0
0.725	5,107 [4,151, 5,707]	5,041 [4,101, 5,623]	6,480 [5,408, 7,167]	100.0	100.0	100.0
0.750	4,623 [3,758, 5,164]	4,563 [3,713, 5,087]	5,860 [4,895, 6,473]	100.0	100.0	100.0
0.775	4,143 [3,369, 4,625]	4,089 [3,330, 4,555]	5,246 [4,384, 5,788]	100.0	100.0	100.0
0.800	3,667 [2,983, 4,092]	3,619 [2,951, 4,029]	4,639 [3,879, 5,110]	100.0	100.0	100.0
0.825	3,195 [2,602, 3,564]	3,153 [2,575, 3,509]	4,038 [3,377, 4,444]	100.0	100.0	100.0
0.850	2,726 [2,223, 3,040]	2,691 [2,201, 2,994]	3,443 [2,881, 3,785]	100.0	100.0	100.0
0.875	2,262 [1,846, 2,521]	2,233 [1,829, 2,483]	2,854 [2,389, 3,134]	100.0	100.0	100.0
0.900	1,802 [1,471, 2,007]	1,778 [1,457, 1,977]	2,271 [1,902, 2,492]	100.0	100.0	100.0
0.925	1,346 [1,100, 1,498]	1,328 [1,088, 1,475]	1,695 [1,421, 1,859]	100.0	100.0	100.0
0.950	893 [730, 994]	881 [722, 979]	1,124 [943, 1,232]	100.0	100.0	100.0
0.975	444 [364, 494]	438 [359, 487]	559 [469, 612]	100.0	100.0	100.0
1.000	−1 [−2, 0]	−1 [−2, 0]	−1 [−2, 0]	0.0	0.0	0.0

B. Threshold JPY 7.5 million per QALY						
Recurrence HR	Exercise added to chemotherapy plus pembrolizumab, base case	Exercise added to chemotherapy plus pembrolizumab, higher mortality after early recurrence with pembrolizumab	Exercise added to chemotherapy alone	Probability above the fee-schedule-based program cost, %: exercise added to chemotherapy plus pembrolizumab, base case	Same, higher mortality after early recurrence with pembrolizumab	Same, exercise added to chemotherapy alone
0.700	7,672 [6,266, 8,533]	7,610 [6,215, 8,454]	9,761 [8,173, 10,782]	100.0	100.0	100.0
0.725	7,003 [5,722, 7,783]	6,946 [5,672, 7,711]	8,900 [7,455, 9,822]	100.0	100.0	100.0
0.750	6,339 [5,181, 7,042]	6,287 [5,137, 6,976]	8,049 [6,744, 8,868]	100.0	100.0	100.0
0.775	5,680 [4,645, 6,307]	5,634 [4,605, 6,246]	7,206 [6,042, 7,930]	100.0	100.0	100.0
0.800	5,028 [4,114, 5,579]	4,987 [4,078, 5,524]	6,372 [5,349, 7,004]	100.0	100.0	100.0
0.825	4,381 [3,586, 4,859]	4,345 [3,554, 4,810]	5,547 [4,659, 6,090]	100.0	100.0	100.0
0.850	3,739 [3,062, 4,144]	3,708 [3,035, 4,103]	4,730 [3,976, 5,188]	100.0	100.0	100.0
0.875	3,102 [2,542, 3,436]	3,077 [2,520, 3,403]	3,921 [3,298, 4,297]	100.0	100.0	100.0
0.900	2,471 [2,026, 2,736]	2,451 [2,008, 2,709]	3,121 [2,628, 3,416]	100.0	100.0	100.0
0.925	1,846 [1,514, 2,042]	1,830 [1,500, 2,022]	2,328 [1,962, 2,546]	100.0	100.0	100.0
0.950	1,225 [1,005, 1,355]	1,215 [997, 1,342]	1,544 [1,302, 1,686]	100.0	100.0	100.0
0.975	610 [501, 674]	604 [497, 668]	768 [648, 838]	100.0	100.0	100.0
1.000	−1 [−2, 0]	−1 [−2, 0]	−1 [−2, 0]	0.0	0.0	0.0

[a]Modeled cohort: women aged 49 years starting neoadjuvant chemotherapy for stage II or III triple-negative breast cancer in Japan (Table 1). Data of Fig. 2. Cells show the deterministic estimate and the 95% UI. 95% UI: 2.5th and 97.5th percentiles of 1,000 probabilistic sensitivity analysis samples. Costs in JPY 1,000 at fiscal year 2026 prices, discounted at 2% per year over a lifetime horizon. The effect is held for 5 years after randomization and then halves every year. Higher mortality after early recurrence with pembrolizumab: Table 1. Probability above the fee-schedule-based program cost: share of samples in which the maximum affordable cost exceeds the base-case cost of that program (Table 1). HR, hazard ratio; JPY, Japanese yen; QALY, quality-adjusted life year; UI, uncertainty interval.

Supplementary Table S6. Maximum affordable cost of an exercise program per patient by the increase in pathological complete response[a]

A. Threshold JPY 5 million per QALY						
Increase in pCR, percentage points	Exercise added to chemotherapy plus pembrolizumab, base case	Exercise added to chemotherapy plus pembrolizumab, higher mortality after early recurrence with pembrolizumab	Exercise added to chemotherapy alone	Probability above the fee-schedule-based program cost, %: exercise added to chemotherapy plus pembrolizumab, base case	Same, higher mortality after early recurrence with pembrolizumab	Same, exercise added to chemotherapy alone
0	−1 [−2, 0]	−1 [−2, 0]	−1 [−2, 0]	0.0	0.0	0.0
1	389 [238, 570]	385 [233, 566]	345 [95, 533]	99.0	99.0	88.4
2	779 [477, 1,142]	770 [467, 1,132]	690 [190, 1,067]	100.0	100.0	97.4
3	1,169 [716, 1,713]	1,155 [701, 1,699]	1,036 [286, 1,601]	100.0	100.0	98.4
4	1,559 [955, 2,285]	1,541 [935, 2,265]	1,381 [381, 2,135]	100.0	100.0	99.1
5	1,949 [1,194, 2,856]	1,926 [1,169, 2,832]	1,726 [476, 2,668]	100.0	100.0	99.2
6	2,339 [1,432, 3,427]	2,311 [1,403, 3,398]	2,072 [572, 3,202]	100.0	100.0	99.3
7	2,728 [1,671, 3,999]	2,697 [1,637, 3,965]	2,417 [667, 3,736]	100.0	100.0	99.3
8	3,118 [1,910, 4,570]	3,082 [1,871, 4,531]	2,763 [763, 4,270]	100.0	100.0	99.3
9	3,508 [2,149, 5,142]	3,467 [2,105, 5,098]	3,108 [858, 4,803]	100.0	100.0	99.3
10	3,898 [2,388, 5,713]	3,853 [2,339, 5,664]	3,454 [953, 5,337]	100.0	100.0	99.3

B. Threshold JPY 7.5 million per QALY						
Increase in pCR, percentage points	Exercise added to chemotherapy plus pembrolizumab, base case	Exercise added to chemotherapy plus pembrolizumab, higher mortality	Exercise added to chemotherapy alone	Probability above the fee-schedule-based program cost, %: exercise	Same, higher mortality after early recurrence with	Same, exercise added to chemotherapy alone

		after early recurrence with pembrolizumab		added to chemotherapy plus pembrolizumab, base case	pembrolizumab	
0	−1 [−2, 0]	−1 [−2, 0]	−1 [−2, 0]	0.0	0.0	0.0
1	546 [345, 788]	543 [341, 785]	495 [155, 751]	100.0	100.0	96.2
2	1,093 [691, 1,576]	1,086 [683, 1,571]	991 [310, 1,503]	100.0	100.0	99.1
3	1,640 [1,037, 2,364]	1,630 [1,026, 2,356]	1,487 [465, 2,256]	100.0	100.0	99.3
4	2,187 [1,382, 3,152]	2,173 [1,368, 3,142]	1,982 [621, 3,008]	100.0	100.0	99.3
5	2,734 [1,728, 3,941]	2,716 [1,710, 3,927]	2,478 [776, 3,760]	100.0	100.0	99.3
6	3,280 [2,074, 4,729]	3,260 [2,052, 4,713]	2,974 [931, 4,512]	100.0	100.0	99.4
7	3,827 [2,420, 5,517]	3,803 [2,394, 5,498]	3,470 [1,087, 5,265]	100.0	100.0	99.4
8	4,374 [2,766, 6,306]	4,347 [2,736, 6,284]	3,966 [1,242, 6,017]	100.0	100.0	99.6
9	4,921 [3,111, 7,094]	4,890 [3,078, 7,069]	4,461 [1,397, 6,769]	100.0	100.0	99.6
10	5,468 [3,457, 7,882]	5,434 [3,421, 7,855]	4,957 [1,552, 7,521]	100.0	100.0	99.6

[a]Modeled cohort: women aged 49 years starting neoadjuvant chemotherapy for stage II or III triple-negative breast cancer in Japan (Table 1). Exercise increases the proportion with pCR by the amount shown (capped at 100%) and does not change the recurrence HR. Cells show the deterministic estimate and the 95% UI. 95% UI: 2.5th and 97.5th percentiles of 1,000 probabilistic sensitivity analysis samples. Costs in JPY 1,000 at fiscal year 2026 prices, discounted at 2% per year over a lifetime horizon. Probability above the fee-schedule-based program cost: share of samples in which the maximum affordable cost exceeds the base-case cost of that program (Table 1). HR, hazard ratio; JPY, Japanese yen; pCR, pathological complete response; QALY, quality-adjusted life year; UI, uncertainty interval.

Supplementary Table S7. Incremental costs by category for exercise added to chemotherapy plus pembrolizumab at a recurrence hazard ratio of 0.85[a]

Incremental cost, JPY 1,000		
Category	Base case	Higher mortality after early recurrence with pembrolizumab
Exercise program (fee-schedule-based program)	192	192
Musculoskeletal adverse events of exercise	1	1
Pembrolizumab	12	12
Drug administration	0	0
Immune-related adverse events	0	0
Febrile neutropenia	0	0
Follow-up	8	8
Locoregional recurrence	−10	−10
Distant recurrence, first-line treatment	−254	−249
Distant recurrence, second-line and later treatment	−437	−397
End-of-life care	−21	−21
Other (chemotherapy drugs, supportive care, surgery, radiotherapy)	0	0
Total, fee-schedule-based program	−509	−464
Total without the program cost[b]	−702	−656
Total, CHALLENGE-based supervised program	−462	−416
Total, remote low-cost program	−690	−645

[a]Modeled cohort: women aged 49 years starting neoadjuvant chemotherapy for stage II or III triple-negative breast cancer in Japan (Table 1). Exercise added to chemotherapy plus pembrolizumab compared with chemotherapy plus pembrolizumab; negative values are costs avoided. Deterministic estimates. Costs in JPY 1,000 at fiscal year 2026 prices, discounted at 2% per year over a lifetime horizon. The effect is held for 5 years after randomization and then halves every year. Categories may not add to the totals because of rounding. JPY, Japanese yen.
[b]Equals minus the medical costs avoided (Table 3).

Supplementary Table S8. Maximum affordable cost of an exercise program per patient under alternative effect durations[a]

A. Threshold JPY 5 million per QALY						
Recurrence HR	Effect until 1 year after surgery: main comparison	Effect until 1 year after surgery: exercise added to chemotherapy alone	No waning: main comparison	No waning: exercise added to chemotherapy alone	Exercise continued after surgery: main comparison	Exercise continued after surgery: exercise added to chemotherapy alone
0.700	3,613	4,669	6,989	9,120	4,685	6,187
0.725	3,302	4,263	6,378	8,316	4,269	5,635
0.750	2,992	3,860	5,773	7,521	3,857	5,088
0.775	2,684	3,460	5,173	6,734	3,448	4,546
0.800	2,378	3,064	4,578	5,954	3,042	4,010
0.825	2,074	2,670	3,989	5,183	2,639	3,480
0.850	1,772	2,279	3,404	4,420	2,239	2,954
0.875	1,472	1,892	2,824	3,664	1,842	2,434
0.900	1,174	1,508	2,250	2,916	1,449	1,919
0.925	878	1,126	1,680	2,176	1,058	1,409
0.950	583	748	1,115	1,443	670	904
0.975	290	372	555	717	286	404
1.000	−1	−1	−1	−1	−96	−91

B. Threshold JPY 7.5 million per QALY						
Recurrence HR	Effect until 1 year after surgery: main comparison	Effect until 1 year after surgery: exercise added to chemotherapy alone	No waning: main comparison	No waning: exercise added to chemotherapy alone	Exercise continued after surgery: main comparison	Exercise continued after surgery: exercise added to chemotherapy alone
0.700	4,990	6,448	9,535	12,451	6,476	8,546
0.725	4,560	5,888	8,702	11,355	5,905	7,786
0.750	4,132	5,331	7,877	10,269	5,338	7,034
0.775	3,707	4,779	7,058	9,194	4,775	6,289
0.800	3,285	4,232	6,247	8,130	4,217	5,552
0.825	2,865	3,688	5,442	7,078	3,664	4,822

0.850	2,448	3,149	4,645	6,035	3,114	4,099
0.875	2,034	2,614	3,854	5,004	2,569	3,384
0.900	1,622	2,083	3,070	3,982	2,028	2,675
0.925	1,212	1,556	2,292	2,971	1,491	1,973
0.950	805	1,033	1,522	1,971	958	1,278
0.975	401	514	757	980	429	590
1.000	−1	−1	−1	−1	−96	−91

[a]Modeled cohort: women aged 49 years starting neoadjuvant chemotherapy for stage II or III triple-negative breast cancer in Japan (Table 1). Deterministic estimates in JPY 1,000 per patient. Main comparison: exercise added to chemotherapy plus pembrolizumab. Effect until 1 year after surgery: the recurrence HR is held until 1 year after surgery and then halves every year; no waning: held for life; exercise continued after surgery: as the first, with exercise continued for 30 months after surgery at its monthly cost, the effect held until the end of continuation and halving every year thereafter, and the continuation cost counted among medical costs (inputs E04 and E05, Supplementary Table S2). The base-case duration is in Supplementary Table S5. Costs in JPY 1,000 at fiscal year 2026 prices, discounted at 2% per year over a lifetime horizon. HR, hazard ratio; JPY, Japanese yen; QALY, quality-adjusted life year.

Supplementary Table S9. Net benefit of an exercise program for one year of eligible patients in Japan, if exercise reduced the recurrence hazard[a]

A. Net benefit per patient, JPY 1,000			
Recurrence HR	Exercise added to chemotherapy plus pembrolizumab, base case	Exercise added to chemotherapy plus pembrolizumab, higher mortality after early recurrence with pembrolizumab	Exercise added to chemotherapy alone
0.95	701	689	931
0.90	1,610	1,586	2,079
0.85	2,534	2,498	3,250

B. Net benefit for one year of patients, JPY billion						
Recurrence HR	N = 3,451: base case	N = 3,451: higher mortality after early recurrence with pembrolizumab	N = 2,900: base case	N = 2,900: higher mortality after early recurrence with pembrolizumab	N = 4,000: base case	N = 4,000: higher mortality after early recurrence with pembrolizumab
0.95	2.68 [2.22, 2.94]	2.66 [2.21, 2.91]	2.26 [1.87, 2.47]	2.23 [1.85, 2.44]	3.11 [2.58, 3.41]	3.08 [2.56, 3.37]
0.90	6.10 [5.16, 6.61]	6.04 [5.13, 6.55]	5.12 [4.34, 5.56]	5.08 [4.31, 5.50]	7.07 [5.98, 7.67]	7.00 [5.94, 7.59]
0.85	9.57 [8.15, 10.36]	9.49 [8.10, 10.25]	8.05 [6.85, 8.71]	7.98 [6.81, 8.62]	11.10 [9.45, 12.01]	11.00 [9.39, 11.88]

[a]Values are conditional on the recurrence HR shown, not expected values. Net benefit per patient: maximum affordable cost at JPY 5 million per QALY minus the cost of the fee-schedule-based program (Table 1). N: patients with stage II or III triple-negative breast cancer (M0) who received neoadjuvant chemotherapy in 2024, 3,451 in total, of whom 2,294 received neoadjuvant pembrolizumab (main comparison) and 1,157 did not (exercise added to chemotherapy alone); Japanese Breast Cancer Society, National Breast Cancer Registry Report No. 55 (cases diagnosed in 2024, final version) [7]. Sensitivity analyses: N = 2,900 (registrations in 2023) and N = 4,000 (allowing for incomplete registration), with the same share receiving pembrolizumab. Panel B: deterministic estimate and 95% UI. 95% UI: 2.5th and 97.5th percentiles of 1,000 probabilistic sensitivity analysis samples. Costs in JPY 1,000 at fiscal year 2026 prices, discounted at 2% per year over a lifetime horizon. HR, hazard ratio; JPY, Japanese yen; QALY, quality-adjusted life year; UI, uncertainty interval.

Supplementary Table S10. One-way sensitivity analysis: maximum affordable cost for exercise added to chemotherapy plus pembrolizumab and ICER of chemotherapy plus pembrolizumab compared with chemotherapy alone[a]

Inputs that changed either result							
Code	Input	Lower bound	Upper bound	Maximum affordable cost, input at lower bound	Maximum affordable cost, input at upper bound	ICER, input at lower bound	ICER, input at upper bound
A01	Paclitaxel plus carboplatin (12 weeks), drug cost	85,302	113,736 [drug price, lower only]	—	—	2,053	2,053
A02	Doxorubicin or epirubicin plus cyclophosphamide (4 cycles), drug cost	25,122	33,496 [drug price, lower only]	—	—	2,053	2,053
A03	Pembrolizumab, drug cost per dose	299,193	398,924 [drug price, lower only]	2,729	2,726	1,296	2,053
A04	Outpatient drug administration, per visit (fees and tests included)	10,768	18,160	2,727	2,726	2,034	2,063
A05	Supportive care (G-CSF, antiemetics), per cycle	5,820	67,296 [mirrored (assumed)]	—	—	2,052	2,053
A06	Surgery (mix of procedures)	641,441	1,250,000	—	—	2,050	2,054
A07	Postoperative radiotherapy, proportion treated	0.50600	0.90000	—	—	2,052	2,054
A08	Postoperative radiotherapy, cost	405,300	485,300 [mirrored	—	—	2,053	2,053

	per course		(assumed)]				
B01	Immune-related AE management cost: hypothyroidism	19,293	236,000	—	—	2,052	2,061
B02	Immune-related AE management cost: hyperthyroidism	19,428	236,000	—	—	2,052	2,056
B03	Immune-related AE management cost: adrenal insufficiency	118,000	472,000	—	—	2,051	2,056
B04	Immune-related AE management cost: hypophysitis	118,000	472,000	—	—	2,052	2,055
B05	Immune-related AE management cost: pneumonitis	31,628	236,000	—	—	2,053	2,053
B06	Immune-related AE management cost: colitis	20,370	236,000	—	—	2,053	2,053
B07	Immune-related AE management cost: hepatitis	20,370	236,000	—	—	2,053	2,054
B08	Immune-related AE management cost: severe skin reactions	38,057	472,000	—	—	2,049	2,058
B09	Immune-related AE management cost: type 1 diabetes	118,000	472,000	—	—	2,052	2,053
B10	Immune-related	118,000	472,000	—	—	2,052	2,055

	AE management cost: other serious events						
B11	Febrile neutropenia management cost per event	189,000	283,000 [mirrored (assumed)]	—	—	2,052	2,053
B21C	Incidence: hypothyroidism (immune-related)	0.03578	0.08438	—	—	2,053	2,052
B21P	Incidence: hypothyroidism (immune-related)	0.12636	0.17771	—	—	2,052	2,053
B22C	Incidence: hyperthyroidism (immune-related)	0.00726	0.03672	—	—	2,053	2,052
B22P	Incidence: hyperthyroidism (immune-related)	0.03783	0.07037	—	—	2,052	2,053
B23C	Incidence: adrenal insufficiency (immune-related)	0.00000	0.00944 [zero events, fixed in PSA]	—	—	2,053	2,052
B23P	Incidence: adrenal insufficiency (immune-related)	0.01567	0.03917	—	—	2,052	2,054
B24C	Incidence: hypophysitis (immune-related)	0.00007	0.01424	—	—	2,053	2,051
B24P	Incidence: hypophysitis (immune-related)	0.01076	0.03140	—	—	2,052	2,054
B25C	Incidence: pneumonitis (immune-related)	0.00062	0.01845	—	—	2,053	2,052

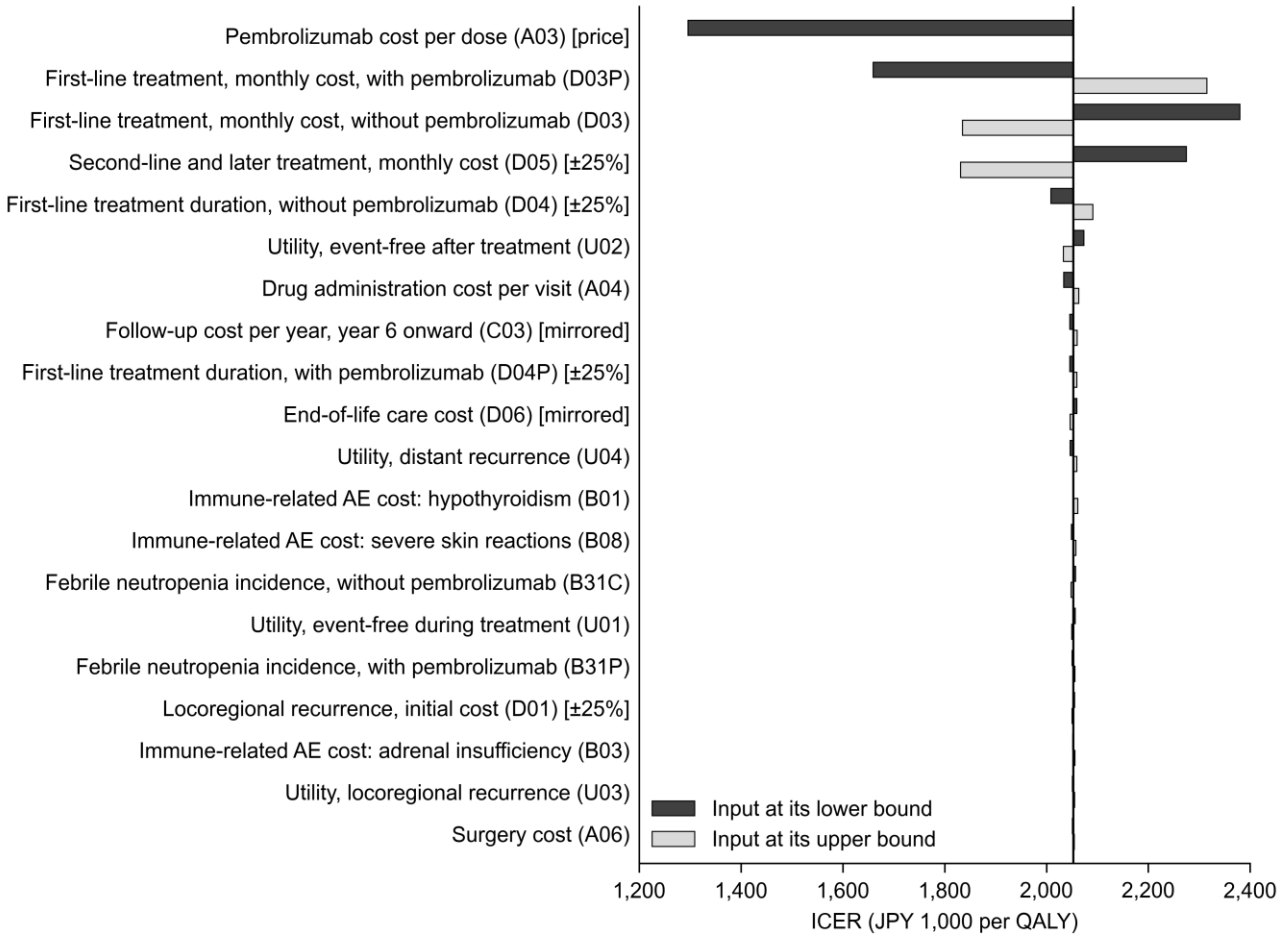
B25P	Incidence: pneumonitis (immune-related)	0.00360	0.01833	—	—	2,053	2,053
B26C	Incidence: colitis (immune-related)	0.00000	0.00944 [zero events, fixed in PSA]	—	—	2,053	2,053
B26P	Incidence: colitis (immune-related)	0.00282	0.01660	—	—	2,053	2,053
B27C	Incidence: hepatitis (immune-related)	0.00000	0.00944 [zero events, fixed in PSA]	—	—	2,053	2,053
B27P	Incidence: hepatitis (immune-related)	0.00527	0.02171	—	—	2,053	2,053
B28C	Incidence: severe skin reactions (immune-related)	0.00007	0.01424	—	—	2,053	2,051
B28P	Incidence: severe skin reactions (immune-related)	0.03349	0.06455	—	—	2,051	2,055
B29C	Incidence: type 1 diabetes (immune-related)	0.00000	0.00944 [zero events, fixed in PSA]	—	—	2,053	2,052
B29P	Incidence: type 1 diabetes (immune-related)	0.00139	0.01303	—	—	2,052	2,054
B30C	Incidence: other serious immune-related AEs	0.00007	0.01424	—	—	2,053	2,051
B30P	Incidence: other serious immune-related AEs	0.01172	0.03297	—	—	2,052	2,054
B31C	Incidence: febrile neutropenia	0.12675	0.20240	—	—	2,057	2,048

B31P	Incidence: febrile neutropenia	0.15737	0.21285	—	—	2,050	2,056
B33C	Grade 3 or higher AEs of six types, events per patient	0.886	1.086	—	—	2,053	2,052
B33P	Grade 3 or higher AEs of six types, events per patient	0.993	1.139	—	—	2,052	2,053
C01	Follow-up cost per year, years 1 to 2	11,660	24,880 [mirrored (assumed)]	2,727	2,726	2,053	2,053
C02	Follow-up cost per year, years 3 to 5	10,120	23,340 [mirrored (assumed)]	2,727	2,726	2,052	2,053
C03	Follow-up cost per year, year 6 onward	9,350	22,570 [mirrored (assumed)]	2,729	2,723	2,046	2,060
D01	Locoregional recurrence, initial treatment cost	916,988	1,528,313 [±25% (assumed)]	2,724	2,729	2,055	2,050
D02	Locoregional recurrence, monthly cost thereafter	2,284	3,806 [±25% (assumed)]	2,726	2,727	2,053	2,053
D03	First-line treatment of distant recurrence, monthly cost	915,446	1,416,257	—	—	2,379	1,835
D03P	First-line treatment of distant recurrence, monthly cost	973,662	1,781,422	2,642	2,783	1,660	2,315
D04	First-line treatment of	7.77	12.96 [±25% (assumed)]	—	—	2,009	2,091

	distant recurrence, mean duration						
D04P	First-line treatment of distant recurrence, mean duration	8.13	13.55 [±25% (assumed)]	2,725	2,728	2,045	2,059
D05	Second-line and later treatment, monthly cost	1,059,715	1,766,192 [±25% (assumed)]	2,617	2,836	2,274	1,831
D06	End-of-life care, last 3 months	956,901	1,283,100 [mirrored (assumed)]	2,723	2,729	2,060	2,046
E06	Musculoskeletal AEs, any grade, with exercise	0.14895	0.22464	2,727	2,726	—	—
E07	Musculoskeletal AEs, any grade, without exercise	0.08731	0.14767	2,726	2,727	—	—
E08	Musculoskeletal AE management cost per event	6,540	11,080 [mirrored (assumed)]	2,727	2,726	—	—
E09	Musculoskeletal AEs, grade 3 or higher, with exercise	0.00145	0.02035	2,727	2,726	—	—
E10	Musculoskeletal AEs, grade 3 or higher, without exercise	0.00005	0.01203	2,726	2,727	—	—
U01	Utility, event-free during treatment	0.881	0.897	2,726	2,726	2,056	2,049
U02	Utility, event-free after treatment	0.877	0.897	2,701	2,751	2,073	2,033
U03	Utility,	0.787	0.873	2,731	2,722	2,051	2,055

	locoregional recurrence						
U04	Utility, distant recurrence or progression during neoadjuvant therapy	0.700	0.770	2,734	2,719	2,046	2,059
U05	Utility decrement per grade 3 or higher AE (12.5 weeks)	0.000	0.059	2,727	2,726	2,052	2,053

[a]Modeled cohort: women aged 49 years starting neoadjuvant chemotherapy for stage II or III triple-negative breast cancer in Japan (Table 1). Data of Fig. 3 and Supplementary Fig. S2. Each input was set to its lower or upper bound with all other inputs at base-case values. Maximum affordable cost: recurrence HR 0.85, threshold JPY 5 million per QALY, JPY 1,000 per patient (base case in Table 3). ICER: JPY 1,000 per QALY (base case in Supplementary Table S4). Bounds are in the units of Supplementary Table S2; brackets mark ranges set by rule (Supplementary Table S2). All 64 inputs that changed either result are listed; a dash means that the input did not change that result. Costs in JPY 1,000 at fiscal year 2026 prices, discounted at 2% per year over a lifetime horizon. HR, hazard ratio; ICER, incremental cost-effectiveness ratio; JPY, Japanese yen; QALY, quality-adjusted life year.



Supplementary Fig. S2. One-way sensitivity analysis of the incremental cost-effectiveness ratio of chemotherapy plus pembrolizumab compared with chemotherapy alone[a]

Bars show the ICER when one input is set to its lower bound (dark bars) or upper bound (light bars) and all other inputs keep their base-case values; the vertical line is the base-case estimate (Supplementary Table S4). The 20 inputs with the widest ranges of the 59 that changed the ICER are shown. Brackets mark ranges set by rule: [±25%], ±25% around the base value; [mirrored], the missing bound mirrored; [price], drug price varied downward only, to 75% of the list price. Codes refer to Supplementary Table S2; all values are listed in Supplementary Table S10.

[a]Modeled cohort: women aged 49 years starting neoadjuvant chemotherapy for stage II or III triple-negative breast cancer in Japan (Table 1). Costs in JPY 1,000 at fiscal year 2026 prices, discounted at 2% per year over a lifetime horizon. AE, adverse event; ICER, incremental cost-effectiveness ratio; JPY, Japanese yen; QALY, quality-adjusted life year.

Supplementary Table S11. Maximum affordable cost, break-even recurrence hazard ratios and the ICER of pembrolizumab under other scenarios[a]

Scenario results							
Scenario	Main comparison: maximum affordable cost at recurrence HR 0.85	Main comparison: recurrence HR (reduction in recurrence hazard, %) at which the maximum affordable cost equals the program cost	Main comparison: recurrence HR (reduction in recurrence hazard, %) at which medical costs avoided equal the program cost	Exercise added to chemotherapy alone: maximum affordable cost at recurrence HR 0.85	Exercise added to chemotherapy alone: recurrence HR (reduction in recurrence hazard, %) at which the maximum affordable cost equals the program cost	Exercise added to chemotherapy alone: recurrence HR (reduction in recurrence hazard, %) at which medical costs avoided equal the program cost	ICER of chemotherapy plus pembrolizumab compared with chemotherapy alone
Utility values from Japanese populations[b]	2,733	0.989 (1.1)	0.958 (4.2)	3,451	0.991 (0.9)	0.966 (3.4)	2,047
Utility value set: United States	2,638	0.989 (1.1)	0.958 (4.2)	3,330	0.991 (0.9)	0.966 (3.4)	2,154
Utility value set: United Kingdom	2,673	0.989 (1.1)	0.958 (4.2)	3,376	0.991 (0.9)	0.966 (3.4)	2,111
Australian reimbursement submission	2,518	0.988 (1.2)	0.958 (4.2)	3,178	0.991 (0.9)	0.966 (3.4)	2,301
Longer survival after progression during neoadjuvant therapy[c]	Same as base case (Table 3)	Same as base case (Table 3)	Same as base case (Table 3)	Same as base case (Table 3)	Same as base case (Table 3)	Same as base case (Table 3)	2,082
Japanese post-operative	2,413	0.988 (1.2)	0.953 (4.7)	2,893	0.990 (1.0)	0.960 (4.0)	Supplementary Table S4

treatment[d]							

[a]Modeled cohort: women aged 49 years starting neoadjuvant chemotherapy for stage II or III triple-negative breast cancer in Japan (Table 1). Deterministic estimates. Main comparison: exercise added to chemotherapy plus pembrolizumab; program cost: the fee-schedule-based program (Table 1). Maximum affordable cost in JPY 1,000 per patient at JPY 5 million per QALY; ICER in JPY 1,000 per QALY. The value in parentheses is the reduction in the recurrence hazard, 100 × (1 − HR), in %. Base-case values are in Table 3 and Supplementary Table S4. Costs in JPY 1,000 at fiscal year 2026 prices, discounted at 2% per year over a lifetime horizon. HR, hazard ratio; ICER, incremental cost-effectiveness ratio; JPY, Japanese yen; QALY, quality-adjusted life year.

[b]Event-free survival during and after treatment 0.89 (EQ-5D-5L with the Japanese value set, a Japanese single-center cohort with primary breast cancer); distant recurrence and progression during neoadjuvant therapy 0.742 (taxane arm of the SELECT BC trial, Japanese value set); no values from Japanese populations were found for locoregional recurrence or the decrement for grade 3 or higher adverse events, which keep their base-case values.

[c]Median survival of 26 months after progression during neoadjuvant therapy instead of the early-recurrence distribution (Supplementary Table S2); this scenario did not reproduce trial overall survival in years 1 to 3. Progression during neoadjuvant therapy occurs before surgery and is the same with and without exercise, whereas the exercise effect on recurrence acts only after surgery. The maximum affordable costs and break-even hazard ratios for exercise therefore equal the base-case values in Table 3 at the precision shown; this scenario changes only the ICER of pembrolizumab.

[d]Capecitabine or olaparib after surgery for patients without pCR (Table 1).

Supplementary Table S12. Cumulative recurrence 5 and 7 years after randomization without exercise, and its absolute reduction with exercise, by recurrence hazard ratio[a]

Recurrence HR	Chemotherapy plus pembrolizumab				Chemotherapy alone			
	Distant recurrence at 5 years	Distant recurrence at 7 years	Any recurrence at 5 years	Any recurrence at 7 years	Distant recurrence at 5 years	Distant recurrence at 7 years	Any recurrence at 5 years	Any recurrence at 7 years
A. Cumulative proportion of women without exercise, %								
1.00	13.13	15.61	15.55	17.35	18.11	20.86	21.02	22.79
B. Absolute reduction with exercise, percentage points								
0.991	0.10	0.11	0.12	0.12	0.13	0.15	0.15	0.16
0.989	0.12	0.14	0.14	0.15	0.16	0.18	0.19	0.20
0.966	0.38	0.43	0.45	0.47	0.50	0.56	0.58	0.61
0.958	0.47	0.53	0.55	0.58	0.62	0.69	0.72	0.75
0.95	0.56	0.63	0.66	0.69	0.74	0.83	0.86	0.89
0.85	1.71	1.93	2.01	2.11	2.28	2.53	2.63	2.74
0.72	3.26	3.68	3.84	4.03	4.36	4.85	5.04	5.24

[a]Values are from the deterministic base case, with all inputs at their base values, the recurrence hazard ratio held for 5 years after randomization and the reduction in the hazard (1 − HR) halving every year thereafter. Cumulative recurrence is the share of the modeled cohort (women aged 49 years starting neoadjuvant chemotherapy for stage II or III triple-negative breast cancer, including those with a pathological complete response and those who progressed before surgery) with a first recurrence after surgery by 5 or 7 years after randomization (model months 60 and 84), counted from the monthly transitions into the recurrence states rather than from the occupancy of those states. Distant recurrence counts the first entry into a distant recurrence state, including entry after a locoregional recurrence; any recurrence counts the first locoregional or distant recurrence. Progression before surgery is not counted. The hazard ratio applies to both the locoregional and the distant recurrence transitions. Absolute reduction is the value without exercise minus the value with exercise, in percentage points. HR, hazard ratio.